(22) If A is skew-symmetric of order n and C is a column matrix of order $n \times 1$, then $C^T AC$ is

(1) an identity matrix of order n (2) an identity matrix of order 1

(3) a zero matrix of order 1 (4) an identity matrix of order 2

(23) The matrix A satisfying the equation $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ is

(1) $\begin{bmatrix} 1 & 4 \\ -1 & 0 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & -4 \\ 1 & 0 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix}$ (4) $\begin{bmatrix} 1 & -4 \\ 1 & 1 \end{bmatrix}$

(24) If $A + I = \begin{bmatrix} 3 & -2 \\ 4 & 1 \end{bmatrix}$, then $(A+I)(A-I)$ is equal to

(1) $\begin{bmatrix} -5 & -4 \\ 8 & -9 \end{bmatrix}$ (2) $\begin{bmatrix} -5 & 4 \\ -8 & 9 \end{bmatrix}$ (3) $\begin{bmatrix} 5 & 4 \\ 8 & 9 \end{bmatrix}$ (4) $\begin{bmatrix} -5 & -4 \\ -8 & -9 \end{bmatrix}$

(25) Let A and B be two symmetric matrices of same order. Then which one of the following statement is not true?

(1) $A + B$ is a symmetric matrix (2) AB is a symmetric matrix

(3) $AB = (BA)^T$ (4) $A^T B = AB^T$

SUMMARY

In this chapter we have acquired the knowledge of

- A matrix is a rectangular array of real numbers or real functions on $\mathbb{R}$ or complex numbers.
- A matrix having m rows and n columns, then the order of the matrix is $m \times n$.
- A matrix $A = [a_{ij}]_{m \times n}$ is said to be a
  square matrix if $m = n$
  row matrix if $m = 1$
  column matrix if $n = 1$
  zero matrix if $a_{ij} = 0 \; \forall \; i$ and j
  diagonal matrix if $m = n$ and $a_{ij} = 0 \; \forall \; i \neq j$
  scalar matrix if $m = n$ and $a_{ij} = 0 \; \forall \; i \neq j$ and $a_{ii} = \lambda$ for all i
  unit matrix or identity matrix if $m = n$ and $a_{ij} = 0$ for all $i \neq j$ and $a_{ii} = 1 \; \forall \; i$
  upper triangular matrix if $m = n$ and $a_{ij} = 0 \; \forall \; i > j$
  lower triangular matrix if $m = n$ and $a_{ij} = 0 \; \forall \; i < j$.
- Matrices $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$, are said to be equal if $a_{ij} = b_{ij}$ and $a_{ij} = b_{ij} \; \forall \; i$ and j
- If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$, then $A + B = [c_{ij}]_{m \times n}$, where $c_{ij} = a_{ij} + b_{ij}$
- If $A = [a_{ij}]_{m \times n}$ and λ is a scalar, then $\lambda A = [\lambda a_{ij}]_{m \times n}$

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